

NITHEESH K K
231901035

Experiment No. 11 Connect to the Internet (Android – Kotlin) Aim

To develop an Android application that connects to the Internet and loads a web page using WebView with proper handling.

Procedure

1. Open Android Studio
2. Create a new project (Empty Views Activity, Kotlin)
3. Add Internet permission in AndroidManifest.xml
4. Design layout with WebView
5. Enable JavaScript in WebView
6. Load a website using URL
7. Handle navigation and errors
8. Run and test the application

AndroidManifest.xml

```
<uses-permission android:name="android.permission.INTERNET"/>
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"    android:layout_height="match_parent"
    android:orientation="vertical">
```

```
    <WebView
        android:id="@+id/webView"
        android:layout_width="match_parent"
        android:layout_height="match_parent"/>
```

```
</LinearLayout> MainActivity.kt
```

```
package com.example.internetconnectapp
```

```
import android.os.Bundle
import android.webkit.WebResourceError
import android.webkit.WebResourceRequest
import android.webkit.WebView import
import android.webkit.WebViewClient import
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity
```

```
class MainActivity : AppCompatActivity() {
```

```
    lateinit var webView: WebView
```

```

        override fun onCreate(savedInstanceState: Bundle?)
        { super.onCreate(savedInstanceState)
            setContentView(R.layout.activity_main)

            webView = findViewById(R.id.webView)

            webView.settings.javaScriptEnabled = true

            webView.webViewClient = object : WebViewClient() {

                override fun shouldOverrideUrlLoading(
                    view: WebView?,
                    request: WebResourceRequest?
                ): Boolean {
                    view?.loadUrl(request?.url.toString())
                    return true
                }

                override fun
                onReceivedError( view: WebView?,
                    request: WebResourceRequest?,
                    error: WebResourceError?
                ) {
                    Toast.makeText( this@MainA
                    ctivity, "Failed
                    to load page",
                        Toast.LENGTH_SHORT
                    ).show()
                }
            }

            webView.loadUrl("https://www.google.com")
        }

        // Handle back navigation
        override fun onBackPressed()
        { if (webView.canGoBack()) {
            webView.goBack()
        } else {
            super.onBackPressed()
        }
    }
}

```

Result

- The application successfully connects to the Internet
- Loads a webpage inside WebView
- Handles navigation and errors

- Allows back navigation within the WebView

